Oriented Boundary Sums, Polynomial Remainders, and Congruence Classes

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§1.1 Three Computations, Three Structures

A coordinate-area formula, a polynomial remainder problem, and a modular-exponent problem can appear on the same exercise sheet without sharing one master theorem. What they do share is a style of simplification: each calculation throws away information that the question does not need.

For a polygon, the relevant data are the oriented contributions of its edges. For a polynomial modulo $x - a$, all multiples of $x - a$ become invisible and only the value at a remains. For an integer modulo m, numbers differing by a multiple of m are treated as the same residue class. The useful lesson is therefore not that every "remainder" is secretly identical, but that one should identify the map that preserves exactly the requested invariant.

§1.2 Determinants as Signed Area

For vectors

$$p = (x_1, y_1), \qquad q = (x_2, y_2),$$

define

$$\det(p, q) = x_1 y_2 - x_2 y_1.$$

Its absolute value is the area of the parallelogram spanned by p, q, while its sign records orientation. Thus the oriented area of the triangle with vertices $0, p, q$ is

$$\frac{1}{2}\det(p, q).$$

Let a polygon have cyclically ordered vertices

$$p_i = (x_i, y_i), \qquad i = 1, \dots, n, \qquad p_{n+1} = p_1.$$

Triangulating from the origin suggests the signed sum

$$\frac{1}{2}\sum_{i=1}^{n} \det(p_i, p_{i+1}).$$

If the origin lies outside the polygon, some triangles point with the opposite orientation. This is not a defect: the negative pieces cancel the extra positive pieces, leaving the polygon's oriented area.

§1.3 Shoelace as a Boundary Integral

Green's theorem gives an origin-free expression for the oriented area of a positively oriented region D:

$$\mathrm{Area}(D) = \frac{1}{2}\int_{\partial D}(x\,dy - y\,dx).$$

For a polygon, parametrize the edge from p_i to p_{i+1} by

$$r_i(t) = p_i + t(p_{i+1} - p_i), \qquad 0 \le t \le 1.$$

Writing $r_i(t) = (x(t), y(t))$, a direct calculation gives

$$\begin{aligned}\frac{1}{2}\int_{p_i}^{p_{i+1}}(x\,dy - y\,dx) &= \frac{1}{2}\int_0^1 \left(x(t)y'(t) - y(t)x'(t)\right)dt \\ &= \frac{1}{2}(x_i y_{i+1} - x_{i+1}y_i).\end{aligned}$$

Summing over the edges yields

$$\boxed{\mathrm{Area}(D) = \frac{1}{2}\left|\sum_{i=1}^{n}(x_i y_{i+1} - x_{i+1}y_i)\right|.}$$

The familiar crossing columns are therefore a mnemonic for a discrete line integral, not the reason the formula is true.

§1.4 Translation and Orientation Checks

The shoelace sum should not depend on where the coordinate origin is placed. Translate every vertex by a fixed vector t. Then

$$\det(p_i + t, p_{i+1} + t) = \det(p_i, p_{i+1}) + \det(t, p_{i+1} - p_i).$$

After summing around the closed polygon, the extra term vanishes because

$$\sum_{i=1}^{n}(p_{i+1} - p_i) = 0.$$

Thus the signed area is translation invariant even though each individual determinant uses position vectors from the origin.

Reversing the cyclic order changes every edge orientation and therefore changes the sign of the sum. The absolute value in the ordinary shoelace formula forgets this orientation. For a self-intersecting closed polygonal path, the same sum computes signed area with winding multiplicity; overlapping regions may cancel rather than simply add.

§1.5 A complete polygon calculation

Take the vertices

$$(3,4),\ (5,11),\ (12,8),\ (9,5),\ (5,6)$$

in the displayed cyclic order. The signed double area is

$$\begin{aligned}2A = {} & 3\cdot 11 - 5\cdot 4 + 5\cdot 8 - 12\cdot 11 \\ & + 12\cdot 5 - 9\cdot 8 + 9\cdot 6 - 5\cdot 5 \\ & + 5\cdot 4 - 3\cdot 6 \\ = {} & -60.\end{aligned}$$

Hence

$$A = 30.$$

The negative sign only says that the listed orientation is clockwise. Reversing the order would give $+60$ without changing the ordinary area.

§1.6 Evaluation removes every multiple of $x - a$

Let k be a field and $f \in k[x]$. Polynomial division gives unique $q \in k[x]$ and $r \in k$ such that

$$f(x) = (x - a)q(x) + r.$$

Substituting $x = a$ kills the divisible part:

$$f(a) = r.$$

Thus the remainder theorem is the statement

$$f(x) \equiv f(a) \pmod{x - a}.$$

This is naturally encoded by the evaluation homomorphism

$$\mathrm{ev}_a : k[x] \longrightarrow k, \qquad f \longmapsto f(a).$$

Its kernel is the ideal $(x - a)$, so the first isomorphism theorem gives

$$k[x]/(x - a) \cong k.$$

The quotient notation is useful because it says exactly what has been discarded: two polynomials become equal when their difference is divisible by $x - a$.

§1.7 Remainders as Interpolation Data

Suppose $f(x)$ leaves remainder 99 modulo $x - 19$ and remainder 19 modulo $x - 99$. The remainder modulo

$$(x - 19)(x - 99)$$

has degree at most one, so write

$$R(x) = ux + v.$$

It must satisfy

$$R(19) = 99, \qquad R(99) = 19.$$

Hence

$$19u + v = 99, \qquad 99u + v = 19,$$

and therefore

$$u = -1, \qquad v = 118.$$

Thus

$$\boxed{R(x) = 118 - x.}$$

This is not a separate trick from interpolation. A polynomial of degree at most one is uniquely determined by its values at two distinct points. In Lagrange form,

$$R(x) = 99\frac{x - 99}{19 - 99} + 19\frac{x - 19}{99 - 19},$$

which simplifies to the same answer.

§1.8 Polynomial CRT and Uniqueness

Because $19 \neq 99$, the ideals $(x-19)$ and $(x-99)$ are comaximal. The Chinese remainder theorem gives

$$k[x]/((x-19)(x-99)) \cong k[x]/(x-19) \times k[x]/(x-99).$$

Using evaluation on each factor,

$$k[x]/((x-19)(x-99)) \cong k \times k.$$

A residue class modulo the quadratic is therefore exactly a pair of values

$$(f(19), f(99)).$$

The linear remainder found above is the unique degree-<2 representative of the pair $(99, 19)$.

More generally, for distinct $a_1, \ldots, a_n$,

$$k[x]/\prod_{i=1}^{n}(x-a_i) \cong k^n,$$

and the inverse map is Lagrange interpolation. The elementary remainder theorem has grown into a structural statement without changing the computation it was designed to perform.

§1.9 Repeated Factors and Hermite Data

Evaluation modulo $x-a$ forgets everything except $f(a)$. A natural question is what remains visible modulo a higher power $(x-a)^2$. Write the exact polynomial Taylor expansion

$$f(x) = f(a) + f'(a)(x-a) + (x-a)^2 q(x),$$

where $q \in \mathbb{Q}[x]$. Then

$$\boxed{f(x) \equiv f(a) + f'(a)(x-a) \pmod{(x-a)^2}.}$$

Two polynomials agree modulo $(x-a)^2$ precisely when their values *and first derivatives* agree at a. Repeated linear factors remember a little more local information than an ordinary point evaluation.

This observation extends the preceding Chinese-remainder example. Suppose we wish to find a polynomial R of degree less than three such that

$$R(1) = 2, \qquad R'(1) = 3, \qquad R(-1) = 4.$$

Start with the first-order Taylor data at 1:

$$R(x) = 2 + 3(x-1) + c(x-1)^2.$$

The remaining value condition gives

$$4 = R(-1) = 2 - 6 + 4c, \qquad c = 2.$$

Thus

$$\boxed{R(x) = 2x^2 - x + 1.}$$

Check directly that $R(1) = 2$, $R'(1) = 4 - 1 = 3$, and $R(-1) = 4$. The calculation has not secretly used three unrelated equations: it is the polynomial Chinese remainder theorem for the coprime factors $(x-1)^2$ and $(x+1)$,

$$\mathbb{Q}[x]/((x-1)^2(x+1)) \cong \mathbb{Q}[x]/(x-1)^2 \times \mathbb{Q}[x]/(x+1).$$

The first component is the pair of numbers $(f(1), f'(1))$, and the second component is $f(-1)$. The total degree bound is $2 + 1 = 3$, matching the three scalar data.

Why is this solution unique? If R and S both have the three prescribed data, their difference is divisible by $(x-1)^2$ and by $(x+1)$. Since these factors are coprime, their product divides $R - S$. But $\deg(R - S) < 3$; therefore $R = S$. This proves uniqueness without a determinant calculation.

§1.10 Infinitesimals in Quotient Rings

There is a particularly memorable way to interpret the first component. Introduce a symbol ε with $\varepsilon^2 = 0$, but *do not* assume $\varepsilon = 0$. The ring

$$\mathbb{Q}[\varepsilon]/(\varepsilon^2)$$

is called the ring of dual numbers. The substitution $x = 1 + \varepsilon$ makes every quadratic and higher term disappear:

$$f(1 + \varepsilon) = f(1) + f'(1)\varepsilon.$$

For instance,

$$(1 + \varepsilon)^5 = 1 + 5\varepsilon + 10\varepsilon^2 + \cdots = 1 + 5\varepsilon.$$

It is easy to verify that

$$\mathbb{Q}[x]/(x-1)^2 \cong \mathbb{Q}[\varepsilon]/(\varepsilon^2)$$

by sending the class of $x-1$ to ε. More generally, the quotient modulo $(x-a)^k$ remembers the first k coefficients of the Taylor expansion of f at a (over a characteristic-zero field).

This is the algebraic germ of a derivative, or a *jet*. We have not introduced a real number smaller than every positive real number: ε lives in a different ring and satisfies $\varepsilon^2 = 0$. Nevertheless, first-order calculus becomes exact arithmetic there. The same distinction between a point value and its first-order neighborhood later appears in tangent vectors, differential geometry, and automatic differentiation.[1]

§1.11 Integer congruence is another quotient arithmetic

For integers, write

$$a \equiv b \pmod{m}$$

when $m \mid (a - b)$. Arithmetic then takes place in the quotient ring

$$\mathbb{Z}/m\mathbb{Z}.$$

Addition and multiplication descend because replacing an integer by another representative of the same class does not change the resulting class.

[1] For Hermite interpolation viewed through Taylor data and polynomial congruences, see Bossoney and Troyanov, *A Direct Approach to Hermite Interpolation*, Sections 1–3.

Division is more delicate. From

$$ca \equiv cb \pmod{m}$$

one may cancel c only when its residue class is a unit, equivalently when

$$\gcd(c, m) = 1.$$

For example,

$$2x \equiv 2 \pmod{6}$$

has the two solutions

$$x \equiv 1, 4 \pmod{6}.$$

Cancelling 2 would incorrectly discard one of them. Quotient arithmetic preserves addition and multiplication automatically, but inverse operations require invertibility.

Small residue sets give quick impossibility tests. Squares satisfy

$$n^2 \equiv 0, 1 \pmod{3}$$

and

$$n^2 \equiv 0, 1, 4 \pmod{8}.$$

A proposed square lying outside these lists is impossible before any large-number calculation begins.

§1.12 Finite unit groups turn huge powers into short cycles

If p is prime and $p \nmid a$, Fermat's theorem says

$$a^{p-1} \equiv 1 \pmod{p}.$$

The reason is that the nonzero residue classes form the finite group

$$(\mathbb{Z}/p\mathbb{Z})^\times$$

of order $p-1$. Lagrange's theorem makes the exponent of every element divide the group order.

To compute

$$2^{1000} \pmod{13},$$

use

$$2^{12} \equiv 1 \pmod{13}$$

and reduce the exponent:

$$1000 = 83 \cdot 12 + 4.$$

Therefore

$$2^{1000} \equiv 2^4 = 16 \equiv 3 \pmod{13}.$$

The quotient ring has retained precisely the residue information while discarding the enormous integer itself.

The three parts of the chapter now end differently. The polygon formula converts a closed boundary into oriented area. Polynomial evaluation converts divisibility by linear factors into interpolation data. Modular arithmetic converts integers into residue classes and exposes finite cycles. Their common practical habit is to choose the correct invariant map before expanding the calculation.